**7** Here are the first 7 terms of a quadratic sequence.

3 6 11 18 27 38 51

First differences: 3,5,7,9,11,13.

(a) Find an expression, in terms of $n$, for the $n$th term of this sequence.

Second differences: 2,2,2,2,2, so start with $n^2$.

**1 mark**
Second difference2

$n^2$: 1,4,9,16,25,36,49; add2 to every term.

**1 mark**
$n^2+2$

nth term = $n^2 + 2$.

**(2)**

(b) Find the 50th term of this sequence.

50th term = $50^2 + 2 = 2500 + 2 = 2502$.

**1 mark**
2502

**(1)**

**(Total for Question 7 is 3 marks)**